GITAM Deemed to be University

Artificial Intelligence and Machine Learning | Course Code: AI201

Artificial Intelligence & Search Strategies - Comprehensive Syllabus Lec

Syllabus Focus: AI201 Full Syllabus Comprehensive Lecture Notes Units 1-5

Summary: Complete syllabus lecture notes covering Artificial Intelligence & Search Strategies.

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Intelligent Agents & State Space Search

" Syllabus Scope: Agent architectures, PEAS definitions, Breadth-First, Uniform Cost, Depth-First, and It

Unit 2: Informed Heuristic Search & Optimality

" Syllabus Scope: Greedy best-first search, A* search algorithm, Admissible and consistent heuristics, Me

AI201: Artificial Intelligence & Search Strategies - Advanced Topics & Assessment

Academic Track: Artificial Intelligence and Machine Learning - Units 3, 4, 5 & Examination Matrix

---

II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Adversarial Search & Games

" Game theory formulation, Minimax decision algorithm, Alpha-Beta pruning optimality, Evaluation functions for

Unit 4: Constraint Satisfaction Problems CSP

" Constraint propagation, AC-3 algorithm, Backtracking with Minimum Remaining Values MRV heuristic, Forward ch

Unit 5: Knowledge Representation & Logic

" Propositional logic, First-order predicate logic, Inference rules, Forward and backward chaining, Resolution

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

1. Prove that A* search is optimal when the heuristic function hn is admissible for tree search and consi
2. Apply Alpha-Beta pruning on a specified 4-ply minimax game tree and show the branches that are pruned.
3. Solve the 4-Queens Constraint Satisfaction Problem using backtracking search and arc-consistency AC-3

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* Stuart Russell, Peter Norvig - Artificial Intelligence: A Modern Approach Pearson 4th Ed
* Elaine Rich, Kevin Knight - Artificial Intelligence McGraw-Hill

Authorized NoteVault Educational Document " Verified for Registered University Students